FLIGHT CONTROLLER

ARKV6X

Triple-IMU FMUv6X autopilot module on the Pixhawk Autopilot Bus

The USA-built ARKV6X flight controller is based on the FMUv6X and Pixhawk Autopilot Bus (PAB) open source standards. An STM32H743 processor reads three time-synchronised IMUs, each on its own SPI bus and its own switched 3.3 V rail, making data averaging, voting and filtering possible rather than leaving the estimator with a single point of failure.

Conforming to the PAB form factor, it drops into any PAB-compatible carrier — the ARK PAB Carrier, ARK Jetson PAB Carrier or ARK Jetson PAB V3 — so one autopilot module moves between airframes and compute payloads.



KEY FEATURES

- Three synchronised IMUs, each independently powered
  - Standard: 2 × TDK InvenSense ICM-42688-P (SPI2, SPI3) and 1 × IIM-42652 industrial IMU (SPI1)
  - Extended Range: 3 × TDK InvenSense IIM-42653 industrial IMU (SPI1–3), ±32 g and ±4000 °/s
- Bosch BMP390 barometer and BMM150 magnetometer on internal I2C4
- Four independently switched 3.3 V sensor rails — one per sensor domain
- Cypress FM25V02A 256 kbit FRAM for non-volatile parameter storage
- 1 W (25 Ω) sensor heater under PWM closed-loop control
- Supercapacitor-backed real-time clock
- microSD logging on a 4-bit SDMMC interface

SPECIFICATIONS

PROCESSOR & MEMORY	
MCU	ST STM32H743IIK6, 176-UFBGA
Flash / RAM	2 MB / 1 MB
FRAM	FM25V02A, 256 kbit, 40 MHz SPI
Oscillators	16.000 MHz + 32.768 kHz ±20 ppm

ELECTRICAL	
Supply voltage	4.5–5.5 V (5 V nominal)
Supply current	500 mA max at 5 V
— main system	300 mA
— sensor heater	200 mA at 5 V (25 Ω, PWM)
Operating temperature	–10 °C to +60 °C

MECHANICAL	
Dimensions	36 × 29 × 5 mm
Weight	5.0 g
Mounting	4 × Ø2.3 mm on 31.4 × 24.4 mm

AT A GLANCE	
Standard	FMUv6X / PAB
MCU	STM32H743IIK6
IMUs	3, synchronised
Baro / Mag	BMP390 / BMM150
Supply	4.5–5.5 V, 500 mA
Operating temp	–10 to +60 °C
Interface	100 + 50 pin PAB
Dimensions	36 × 29 × 5 mm
Weight	5.0 g
Hardware rev	Rev 4.1 / 5.1

COMPLIANCE & SOURCING

- Designed and manufactured in the USA
- NDAA compliant
- Blue UAS Framework listed
- FMUv6X open standard
- Pixhawk Autopilot Bus DS-010 standard

ORDERING & RESOURCES

STORE arkelectron.com/product/arkv6x/
arkelectron.com/product/arkv6x-iim-42653/

DOCS docs.arkelectron.com/products/flight-controller/arkv6x

CAD github.com/ARK-Electronics/ark-hardware

BUNDLE ARKV6X Bundle — with ARK PAB Carrier

VARIANTS

	ARKV6X	ARKV6X EXTENDED RANGE
IMU on SPI1	TDK InvenSense IIM-42652	TDK InvenSense IIM-42653
IMUs on SPI2, SPI3	TDK InvenSense ICM-42688-P	TDK InvenSense IIM-42653
Accel full scale	±16 g	±32 g
Gyro full scale	±2000 °/s	±4000 °/s
Hardware revision ID	ARKV6X000	ARKV6X001
Schematic	Rev 4.1	Rev 5.1

Same board, connectors and pinout.

PIXHAWK AUTOPILOT BUS INTERFACE

PAB CONNECTOR	HIROSE PART	SIGNALS CARRIED
PAB X1 (J2)	DF40C-100DP-0.4V, 100 pos	8 × PWM · CAN1 / CAN2 · 8 × UART (see serial map) · I2C1–I2C3 · USB with VBUS sense · PPM and RC input · SWD · safety switch, buzzer, armed · 3 × power-input sense · ADC · 5 V input
PAB X2 (J3)	DF40C-50DP-0.4V, 50 pos	Ethernet RMII with power enable · external SPI6 (2 × CS, reset, 2 × DRDY) · SPIX_SYNC · 17 reserved pins

SENSOR BUS MAP

DEVICE	INTERFACE	ADDRESS / SELECT	FUNCTION
IIM-42652 or IIM-42653 (U9)	SPI1	nCS1	Industrial 6-axis accel / gyro
ICM-42688-P or IIM-42653 (U12)	SPI2	nCS1	6-axis accelerometer / gyroscope
ICM-42688-P or IIM-42653 (U13)	SPI3	nCS1	6-axis accelerometer / gyroscope
BMP390 (U11)	I2C4	0x76	Barometric pressure
BMM150 (U10)	I2C4	0x10	3-axis magnetometer

Where two IMUs are named, the first is on the ARKV6X and the second on the Extended Range.

CARRIER BOARD OPTIONS

ARK CARRIER BOARD	AVIONICS I/O	ADDS
ARK Pixhawk Autopilot Bus Carrier	JST-GH	Full Pixhawk-standard breakout, 100 Mbps Ethernet, dual power
ARK Jetson PAB Carrier	JST-GH	NVIDIA Jetson Orin compute on Pixhawk-standard ports
ARK Jetson PAB V3	Pico-Clasp	Jetson Orin compute, triple power inputs, Ethernet switch
ARK VOXL2 RTK PAB Carrier	Molex	ModalAI VOXL2 compute; Blue UAS Framework listed

ALTERNATIVES IN THE ARK AUTOPILOT LINE

AUTOPILOT	INERTIAL SENSING	POWER INPUT	WHEN TO CHOOSE IT
ARKV6S	1 industrial IMU	5 V via carrier	Cost-optimised, no IMU redundancy
ARK FPV	1 industrial IMU	3–12S direct	Standalone 30.5 mm build, 12 V payload rail

SERIAL REFERENCE & FIRMWARE SUPPORT

SERIAL PORT MAPPING

UART	DEVICE	PORT
USART1	/dev/ttyS0	GPS
USART2	/dev/ttyS1	TELEM3
USART3	/dev/ttyS2	Debug Console
UART4	/dev/ttyS3	UART4 & I2C
UART5	/dev/ttyS4	TELEM2
USART6	/dev/ttyS5	PX4IO/RC
UART7	/dev/ttyS6	TELEM1
UART8	/dev/ttyS7	GPS2

FIRMWARE & SOFTWARE SUPPORT

- PX4 Autopilot — ships by default; ArduPilot supported
- Board-to-board pinout follows the DS-010 Pixhawk Autopilot Bus Standard
- Carrier required. All I/O is presented on the two PAB board-to-board connectors; wiring connectors are provided by the carrier board.

ARK-DS-ARKV6X | Version 1.0, 2026-10-29 | Compiled from the ARKV6X Production schematics Rev 4.1 (13 Dec 2023) and Rev 5.1 Extended Range (19 Jul 2024), the ARKV6X Production bill of materials, the ARKV6X production STEP model, and ARK Electronics published documentation. Temperature ratings per component data sheets; the RTC backup supercapacitor sets both limits. Specifications are subject to change without notice. © 2026 ARK Electronics LLC.